
Neutrinos via Charm Decays in Astrophysical Sources

TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

Fritz Ali Agildere

September 3, 2024

Technical University of Dortmund & Ruhr University Bochum



Particle Physics

- Point A
- Point B
 - Part 1
 - Part 2
- Point C
- Point D



Multimessenger Astronomy

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



Astrophysical Sources

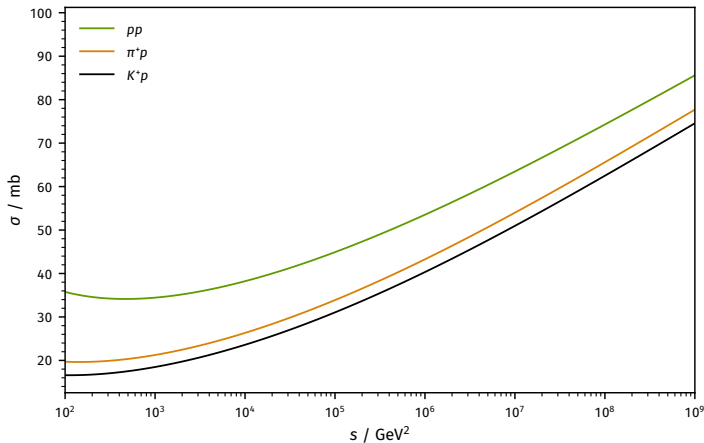
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



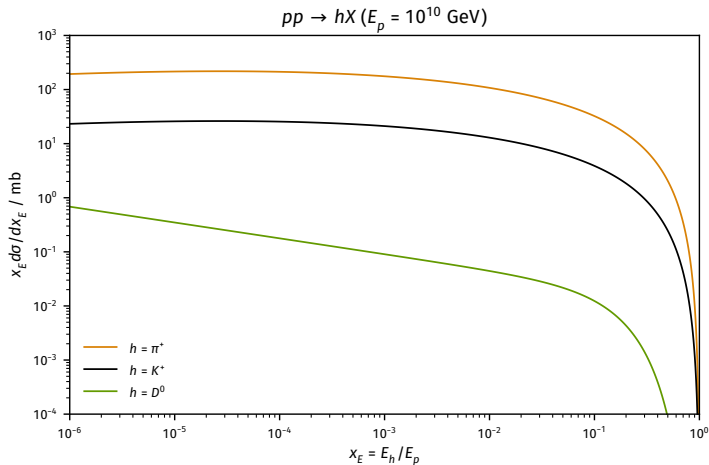
Cross Sections

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Hadron Inelastic Scattering Cross Section



Weighted Energy Differential Cross Section





Spectral Distributions

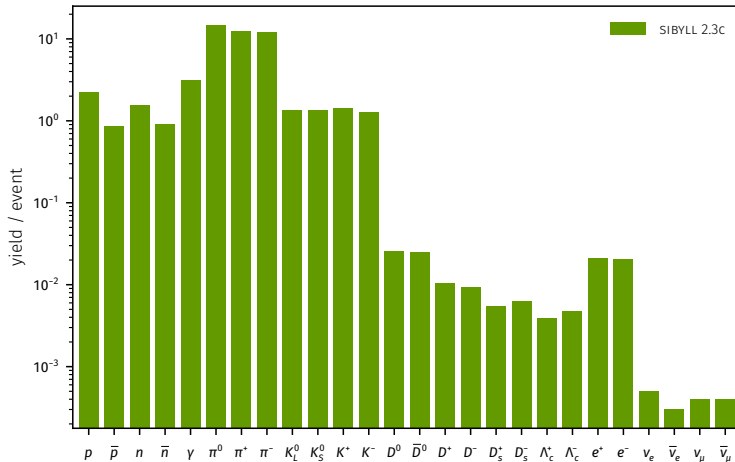
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



Computation

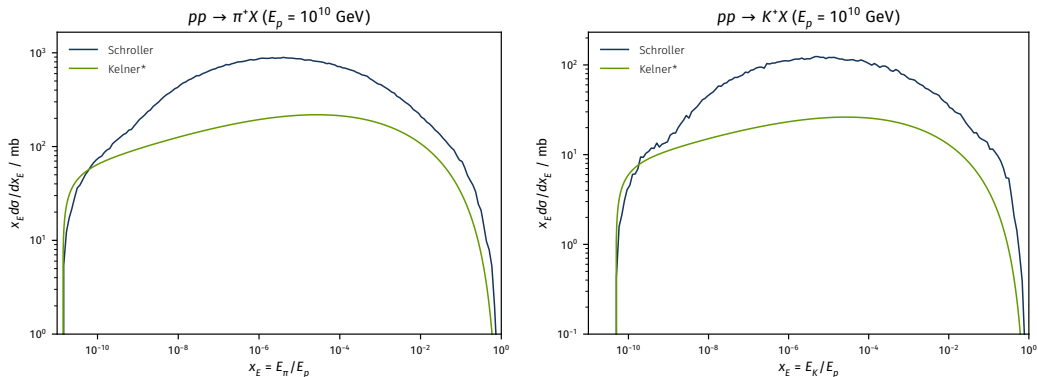
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Event Generation



Data Courtesy
M. Schroller

Weighted Energy Pion & Kaon Differential Cross Section



Derived Parametrization S. R. Kelner et. al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schrollier



Results

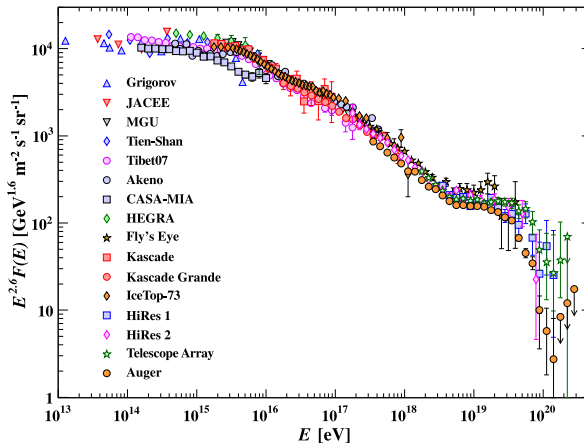
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



Conclusion & Outlook

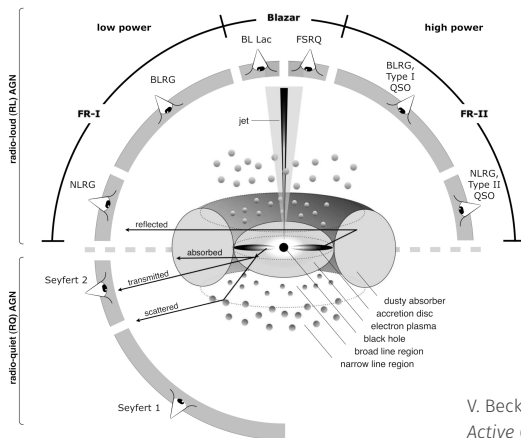
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. [1]

Cosmic Ray Spectrum & Cutoff



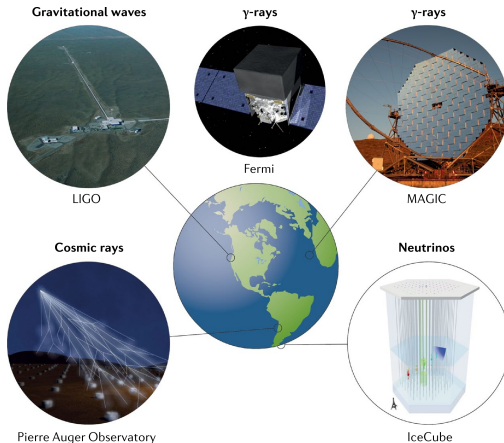
K. A. Olive et. al.
Chin. Phys. C (2014)

Unified AGN Classification Model



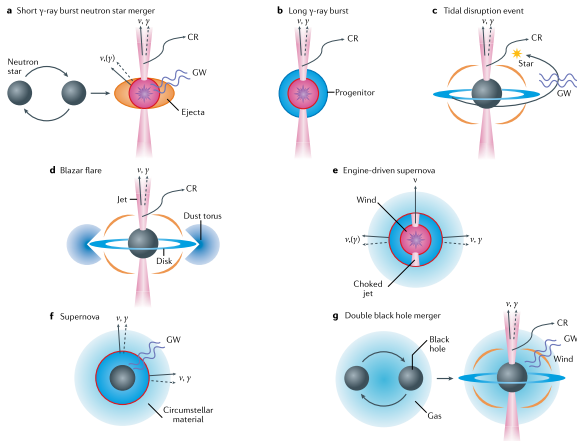
V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)

Detectors



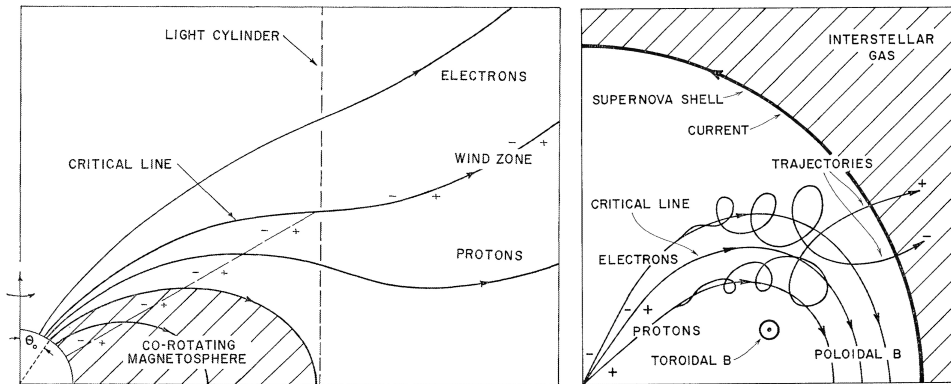
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Sources



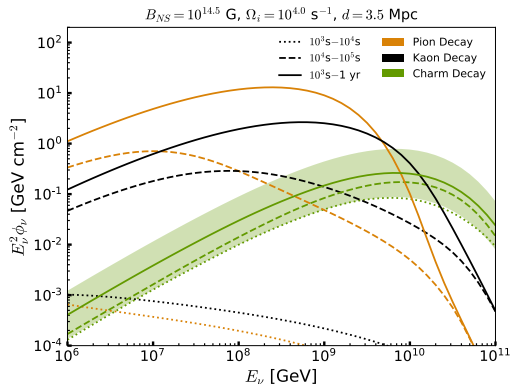
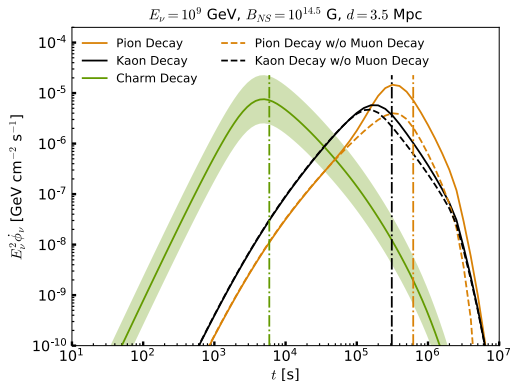
P. Mészáros et. al.
Nat. Rev. Phys. (2019)

Pulsar Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)

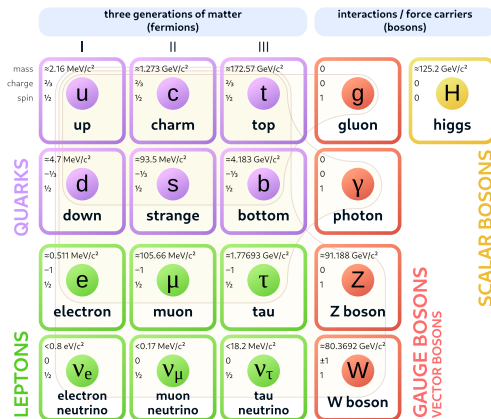
Magnetar Neutrino Flux & Fluence Comparison



J. A. Carpio et. al. Phys. Rev. D (2020)

Elementary Particles

Standard Model of Elementary Particles



Standard Model
Wikimedia Commons (2019)